AI in Thermal-Fluids: Week 16

Supersonic Shape Optimization and Sonic-Boom Physics

From verified CFD to POD surrogates and independently recomputed designs

The learning question. How can we change the shape of a supersonic body to reshape its pressure signature, accelerate the search with machine learning, and still make a scientifically defensible claim after the optimizer has finished?

After Week 16, you should be able to

1. Explain Mach waves, shock formation, near-field pressure signatures, the idealized N-wave, and why near-field peak pressure is not the same quantity as ground loudness.
2. Derive the conversion between C_p and $\Delta p/p_\infty$, and explain why the normalization matters when comparing CFD with sonic-boom reference data.
3. Derive the fixed-volume two-parameter axisymmetric body family used in the computational experiment.
4. State the axisymmetric Euler model, its assumptions, and the boundary/observation quantities used by the SU2 calculations.
5. Distinguish iterative convergence, mesh sensitivity, analytical verification, experimental validation, and machine-learning validation.
6. Explain how POD compresses the pressure waveform and how a small neural network maps geometry to POD coefficients and pressure drag.
7. Use validation data to select a surrogate, quantify waveform/peak/drag errors, and explain why extrapolation is much harder than interpolation.
8. Explain why an optimizer-selected geometry must be recomputed with CFD, then interpret the retained Week 16 design results and their mesh/off-design checks.

Scope. The accepted teaching problem is a fixed-volume body of revolution at zero incidence, $M_\infty = 1.8$, solved with axisymmetric compressible Euler CFD. The design target is the off-body pressure signature at $r/L = 0.5$. The module does *not* calculate atmospheric propagation or PLdB. NASA SEEB-ALR is retained as a reference/failure-analysis extension; the attempted SU2 reproduction did not pass acceptance and is not presented as successful validation.

Reading route.

Sections	Purpose
1–4	Supersonic waves, sonic-boom signatures, pressure measures, propagation
5–8	Geometry, governing equations, numerical model, Taylor–Maccoll verification
9–12	CFD campaign, POD reduction, neural surrogate, error metrics
13–15	Optimization, independent CFD recomputation, off-design robustness
16–18	NASA reference lesson, what failed, and what would be required for ground noise
19	Reproducibility checklist and final takeaways

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1 Mach waves and the Mach cone

The local speed of sound is the speed at which an infinitesimal pressure disturbance propagates relative to the gas. For a calorically perfect gas,

$$a = \sqrt{\gamma RT}. \tag{1}$$

The Mach number compares the body speed U with a:

$$\boxed{M = \frac{U}{a}}. \tag{2}$$

When $M < 1$, acoustic information can propagate ahead of the body. When $M > 1$, the body outruns infinitesimal disturbances. The envelope of emitted wavelets forms a Mach cone. Geometry gives

$$\boxed{\mu = \sin^{-1}\left(\frac{1}{M}\right)}, \tag{3}$$

where μ is the Mach angle. At $M = 1.8$, $\mu \approx 33.75^\circ$.

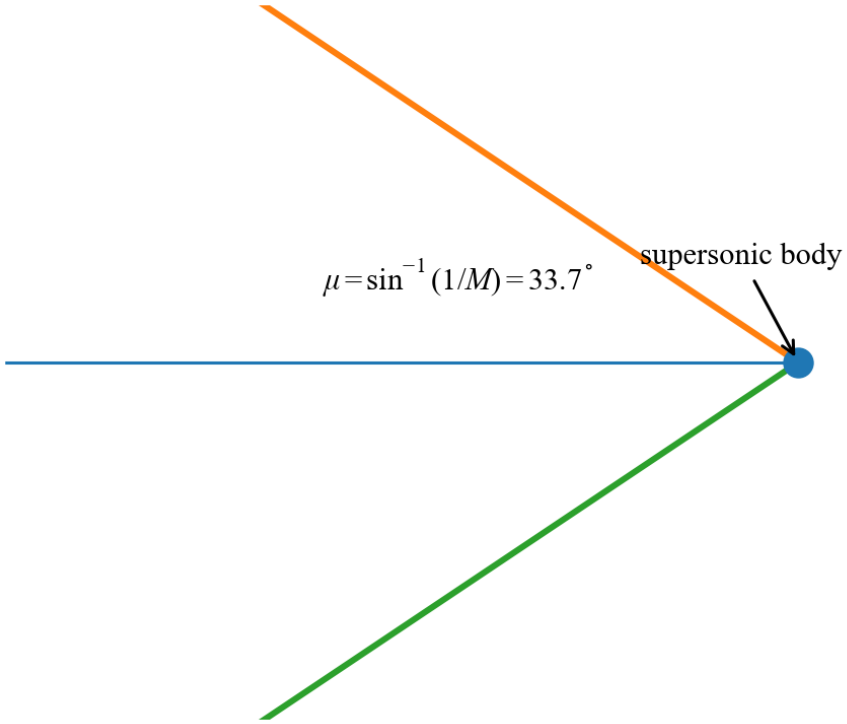


Figure 1: Schematic Mach cone for $M = 1.8$. The angle is set by kinematics; finite-amplitude compression waves can steepen into shocks.

A slender body does not generate one isolated wave. Every change of surface slope and cross-sectional area creates compression or expansion disturbances. Close to the body, individual waves remain distinguishable. Farther away, nonlinear propagation causes compression waves to steepen and disturbances to interact.

Check your understanding. Why does the Mach cone become narrower as M increases? What happens to μ as $M \to 1^+$?

2 From compression waves to a sonic-boom pressure history

A supersonic body creates a spatial pressure pattern. An observer samples that pattern in time as the vehicle passes. The pressure perturbation is

$$p'(t) = p(t) - p_\infty. \tag{4}$$

Compression raises pressure; expansion lowers it. In the far field of a conventional slender configuration, many disturbances may merge into an idealized N-shaped signature: a rapid positive jump, a gradual decrease through ambient pressure, and a final jump back toward ambient.

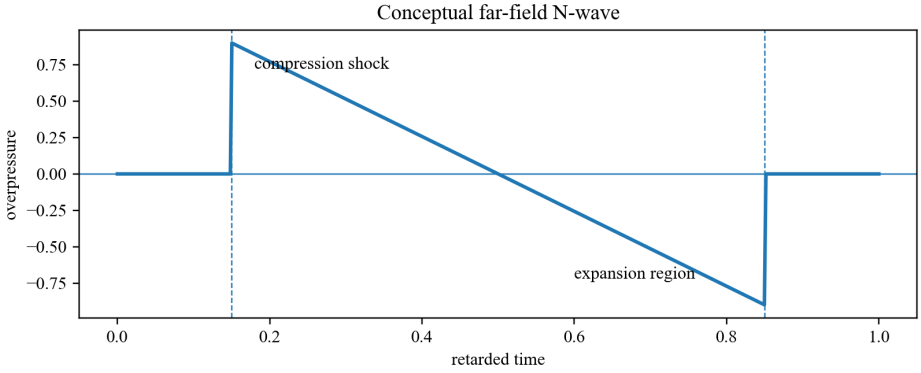


Figure 2: Conceptual N-wave. This is a teaching schematic, not a waveform computed in Week 16.

The two sharp changes are shocks. Their finite rise time and amplitude at the ground depend on propagation distance, geometry, altitude, atmosphere, humidity, winds, and nonlinear effects. A "low-boom" vehicle aims to distribute volume and lift so that the individual near-field disturbances do not coalesce into a strong conventional N-wave.

Key idea. The geometry controls how compression and expansion disturbances are launched. Atmospheric propagation controls how the near-field signature evolves before reaching an observer. These are related but distinct modeling stages.

3 Pressure coefficient and sonic-boom normalization

Aerodynamic CFD commonly reports the pressure coefficient

$$\boxed{C_p = \frac{p - p_\infty}{q_\infty}}, \qquad q_\infty = \frac{1}{2}\rho_\infty U_\infty^2. \tag{5}$$

Using $U_\infty = M_\infty a_\infty$ and $a_\infty^2 = \gamma p_\infty/\rho_\infty$,

$$q_\infty = \frac{1}{2}\rho_\infty M_\infty^2 \frac{\gamma p_\infty}{\rho_\infty} = \frac{1}{2}\gamma p_\infty M_\infty^2, \tag{6}$$

$$\therefore \quad \boxed{\frac{\Delta p}{p_\infty} = \frac{\gamma M_\infty^2}{2} C_p}. \tag{7}$$

At $M_\infty = 1.8$ with $\gamma = 1.4$, the multiplier is 2.268. At the NASA SEEB-ALR condition $M_\infty = 1.6$, it is 1.792.

This conversion is essential. Two curves with the same physical pressure cannot be compared numerically if one is normalized by q_∞ and the other by p_∞.

Check your understanding. If $C_p = 0.03$ at $M = 1.8$, what is $\Delta p/p_\infty$? Why would copying the same number directly onto a NASA $\Delta p/p_\infty$ plot be wrong?

4 Equivalent area, Whitham theory, and propagation: where Week 16 fits

Classical sonic-boom theory connects body shape and lift distribution to an equivalent area distribution. In slender-body theory, the longitudinal evolution of that equivalent area determines the strength and spacing of disturbances. Whitham's F-function reformulates the source distribution so that weak nonlinear propagation can be treated efficiently. The exact full-aircraft construction is beyond the present two-parameter teaching body, but the conceptual chain is important:

$$\begin{aligned}&\text{geometry and lift} \rightarrow \text{equivalent area} \rightarrow F\text{-function}\\ &\quad \rightarrow \text{near-field signature} \rightarrow \text{propagation} \rightarrow \text{ground waveform.}\end{aligned} \tag{8}$$

A frequently used weak-shock propagation model is a generalized Burgers-type equation. In schematic form,

$$\frac{\partial p'}{\partial s} + \beta p' \frac{\partial p'}{\partial \tau} = \delta \frac{\partial^2 p'}{\partial \tau^2} + \mathcal{G}[p'] + \mathcal{A}[p'], \tag{9}$$

where s is propagation distance and τ is retarded time. The nonlinear term steepens compression waves; the diffusion-like term represents absorption; $\mathcal{G}$ represents spreading; and $\mathcal{A}$ can include atmospheric stratification, relaxation, wind, and humidity effects.

Week 16 stops *before* this propagation stage. We optimize a near-field C_p signature, so the strongest defensible statement is about near-field pressure shaping. We do not infer a PLdB value.

5 The fixed-volume body family

We need a shape parameterization that changes the longitudinal distribution without allowing the optimizer to reduce the objective by simply shrinking the body. Let $s = x/L$, $0 \le s \le 1$. Define

$$f(s; a, b) = \sin(\pi s) \exp\left[a(2s-1) + b\cos(2\pi s)\right], \tag{10}$$

and

$$r(s) = cf(s; a, b). \tag{11}$$

For a body of revolution,

$$V = \pi L \int_0^1 r(s)^2 \, \mathrm{d}s = \pi L c^2 \int_0^1 f(s)^2 \, \mathrm{d}s. \tag{12}$$

Solving for c gives the volume-preserving scale factor

$$\boxed{c = \sqrt{\frac{V}{\pi L \int_0^1 f(s)^2 \, \mathrm{d}s}}}. \tag{13}$$

Thus a and b redistribute volume, while c restores the prescribed total volume.

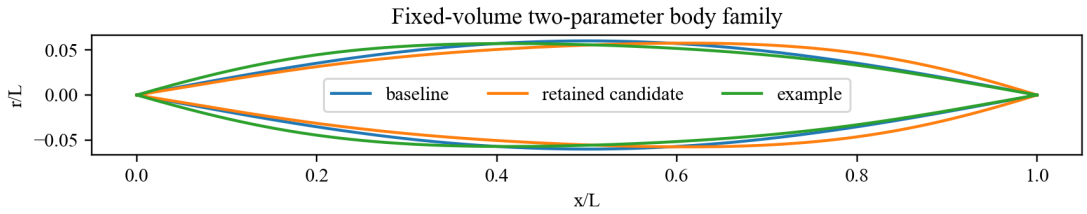


Figure 3: Examples from the fixed-volume two-parameter family. The retained optimizer candidate is geometrically distinct from the baseline but has the same target volume.

The teaching configuration uses $L = 1$ and $V = \pi(0.06)^2/2$. The exported-wall volume audit requires the realized mesh geometry to remain within 0.2% of the target.

6 Axisymmetric compressible Euler equations

The physical body is three-dimensional and axisymmetric. The meridional mesh is a computational representation of the axisymmetric equations; it is not a planar two-dimensional airfoil calculation.

For conservative variables

$$\mathbf{U} = \begin{bmatrix} \rho & \rho u & \rho v & \rho E \end{bmatrix}^T, \tag{14}$$

the axisymmetric Euler equations for no swirl can be written as

$$\boxed{\frac{\partial (r\mathbf{U})}{\partial t} + \frac{\partial (r\mathbf{F}_x)}{\partial x} + \frac{\partial (r\mathbf{F}_r)}{\partial r} = \mathbf{S}_r}, \tag{15}$$

where $\mathbf{S}_r = [0, 0, p, 0]^T$. For radial momentum, multiplying by r and using $\partial_r(rp) = r\partial_r p + p$ gives the pressure source p (not p/r in this weighted form). Uniform stagnant pressure therefore balances exactly. The ideal-gas closure is

$$p = (\gamma - 1)\left[\rho E - \frac{\rho}{2}(u^2 + v^2)\right]. \tag{16}$$

The model is inviscid, so pressure drag is the relevant aerodynamic constraint. It does not include skin friction, viscous heating, turbulence, propulsion, wings, lift, or angle of attack. The freestream is $p_\infty = 101325$ Pa, $T_\infty = 288.15$ K, and $M_\infty = 1.8$. These values define a nondimensional teaching problem; they are not intended as a cruise-altitude atmospheric condition.

7 CFD workflow: Gmsh, SU2, extraction, and acceptance gates

The computational pipeline is

$$(a, b) \rightarrow \text{body profile} \rightarrow \text{Gmsh mesh} \rightarrow \text{SU2 Euler solve} \rightarrow \text{physical/numerical audit} \rightarrow C_p(x, r/L) \text{ extraction.} \tag{17}$$

Pressure signatures are extracted at $r/L = 0.25$, 0.5, and 0.75; the primary design line is $r/L = 0.5$.

For every clean label, acceptance requires finite positive pressure and density, final density residual $\log_{10} R_\rho \leq -9$, residual drop of at least five orders of magnitude, pressure-drag variation below 10^{-4} over the last 100 iterations, maximum total-enthalpy deviation no larger than 10%, and density below 110% of the isentropic stagnation bound. These are rejection gates, not uncertainty bounds.

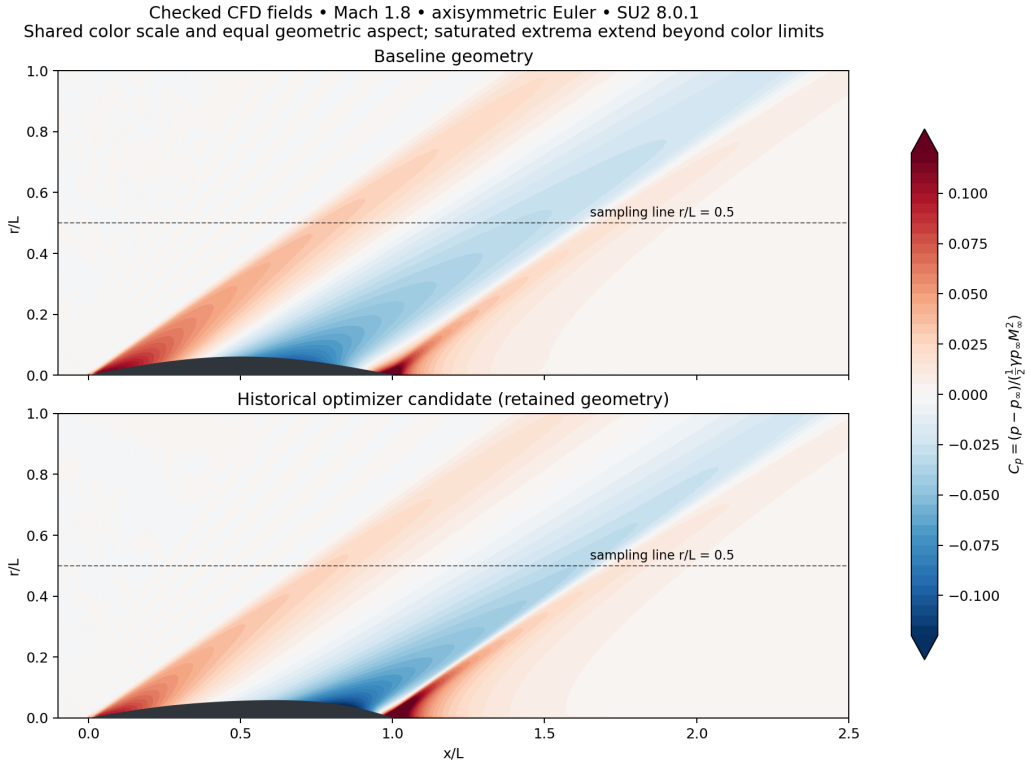


Figure 4: Accepted SU2 8.0.1 pressure fields for the baseline and retained optimizer geometry. The dashed line is the primary sampling location $r/L = 0.5$. Shared color limits make redistribution of the compression/expansion field directly visible.

Check your understanding. Why can a run have a very small algebraic residual and still be rejected physically?

8 Independent verification with Taylor–Maccoll cone flow

Before trusting the teaching-body campaign, we ask whether the numerical method reproduces an independent supersonic solution. Axisymmetric conical flow provides a classical verification problem. An oblique shock turns the freestream toward the cone, and the post-shock flow between shock and surface is governed by the Taylor–Maccoll ordinary differential equations. The shock angle is adjusted until the polar velocity becomes tangent to the cone surface. With velocities normalized by maximum adiabatic speed, define $h = (\gamma - 1)(1 - v_r^2 - v_\theta^2)/2$. The shooting system is

$$\frac{\mathrm{d}v_r}{\mathrm{d}\theta} = v_\theta,$$
$$\frac{\mathrm{d}v_\theta}{\mathrm{d}\theta} = \frac{v_\theta^2 v_r - h(2v_r + v_\theta \cot\theta)}{h - v_\theta^2}.$$

Start at a trial shock angle using oblique-shock jump velocities, integrate toward the cone half-angle, and adjust the shock angle until $v_\theta(\theta_c) = 0$. The notebook executes this ODE and root solve without launching CFD. At $M = 1.8$ and $\theta_c = 7^\circ$, it gives approximately 34.04° shock angle and $C_p = 0.0620$.

The benchmark is independent of the SU2 discretization used for the production data. Therefore agreement tests the numerical chain rather than simply comparing SU2 against itself.

Three SU2 8.0.1 meshes contain approximately 24,000, 96,000, and 216,000 cells. The mean cone-pressure errors are 4.563%, 2.156%, and 1.162%, respectively. The last-two-grid mean-C_p change is 1.006%. All three solutions pass numerical and physical checks; the coarse accuracy gate is intentionally retained as a failure, while the declared refined-family criteria pass.

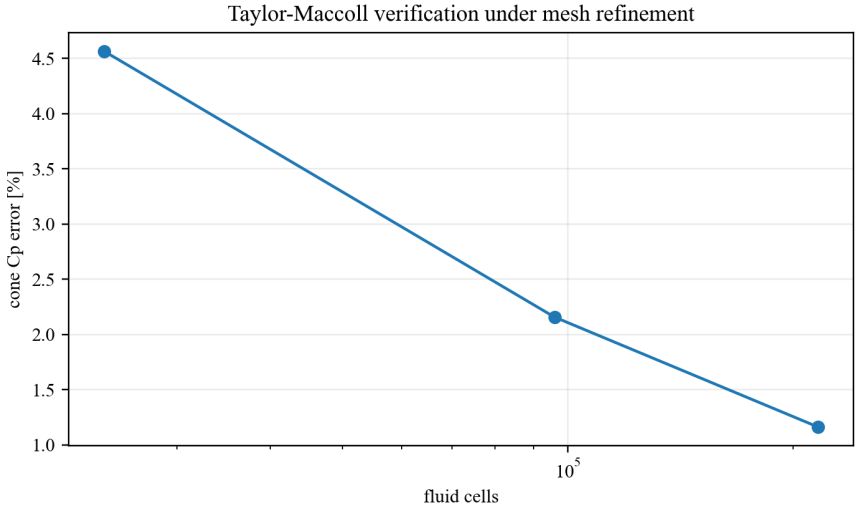


Figure 5: Taylor–Maccoll verification. Refinement reduces the analytical cone-pressure error; the finest result is within the declared 3% criterion.

Key idea. Verification asks whether the equations are solved correctly. Experimental validation asks whether the chosen equations/model describe reality for the intended quantity. They are not interchangeable.

9 The 44-case clean CFD campaign

The final clean-label campaign uses the checksum-pinned official SU2 8.0.1 executable. All 44 original geometries were recomputed and passed the declared numerical and full-field physical gates. The split is frozen:

Split	Train	Validation	Test	Extrapolation
Cases	24	6	8	6

All four groups come from the same two-parameter body family. The test set measures interpolation to withheld geometries. The extrapolation set extends a shape parameter beyond the training range; it is still not a new aircraft topology.

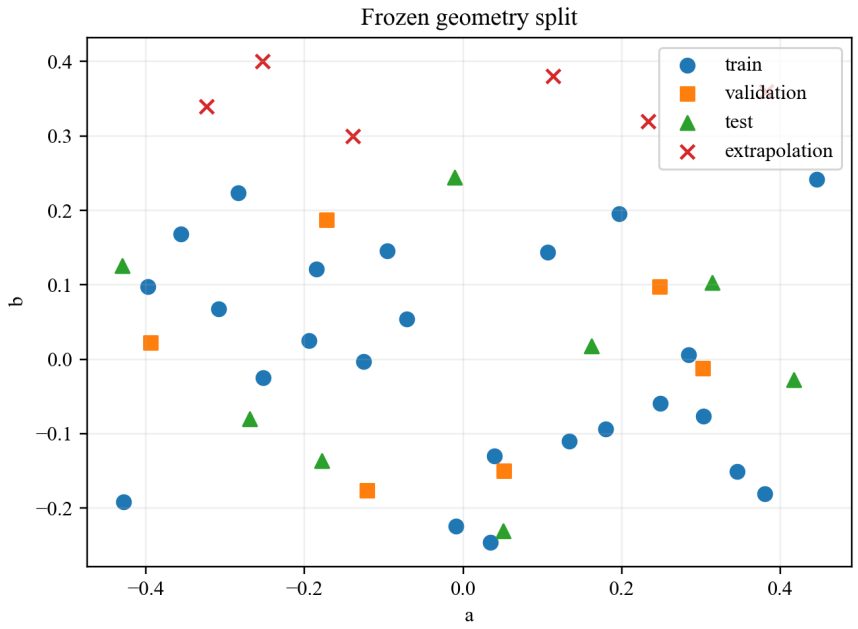


Figure 6: Frozen geometry split in (a, b) space. No validation, test, or extrapolation geometry is used to fit the training-only preprocessing or neural weights.

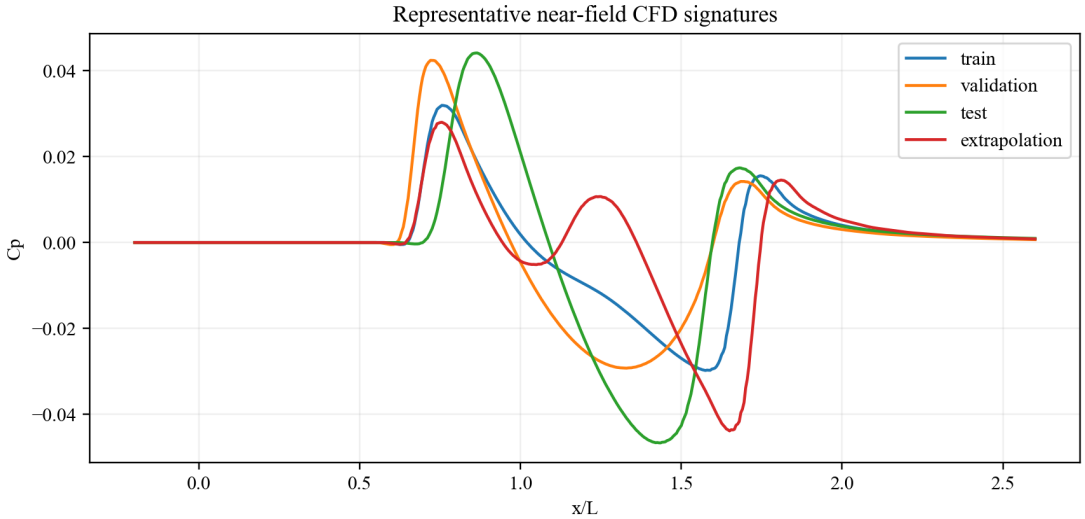


Figure 7: Representative accepted CFD pressure signatures from each split. Shock location and amplitude both change with geometry.

10 POD: compressing a pressure waveform into a few coordinates

Each CFD case contains a vector of pressure samples

$$\mathbf{y}_i = [C_p(x_1), \dots, C_p(x_m)]. \tag{18}$$

Stack the 24 training waveforms into $\mathbf{Y}$ and subtract the training mean $\bar{\mathbf{y}}$. Singular-value decomposition gives

$$\mathbf{Y}' = \mathbf{U}\Sigma\mathbf{V}^T. \tag{19}$$

The rows of $\mathbf{V}^T$ are orthonormal spatial modes $\phi_k(x)$. A waveform is approximated by

$$\boxed{C_p(x;\boldsymbol{\theta}) \approx \bar{C}_p(x) + \sum_{k=1}^{K} z_k(\boldsymbol{\theta})\phi_k(x)}, \tag{20}$$

with $\boldsymbol{\theta} = (a, b)$. The retained energy is

$$E_K = \frac{\sum_{k=1}^{K} \sigma_k^2}{\sum_j \sigma_j^2}. \tag{21}$$

The released clean model uses $K = 12$ full-SVD POD modes. Critically, the mean, basis, input scaling, and output scaling are all fitted on the 24 training cases only.

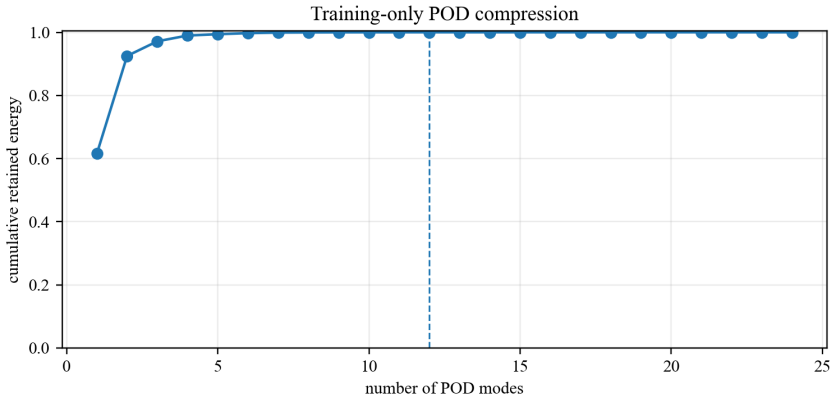


Figure 8: Cumulative POD energy from the training set only.

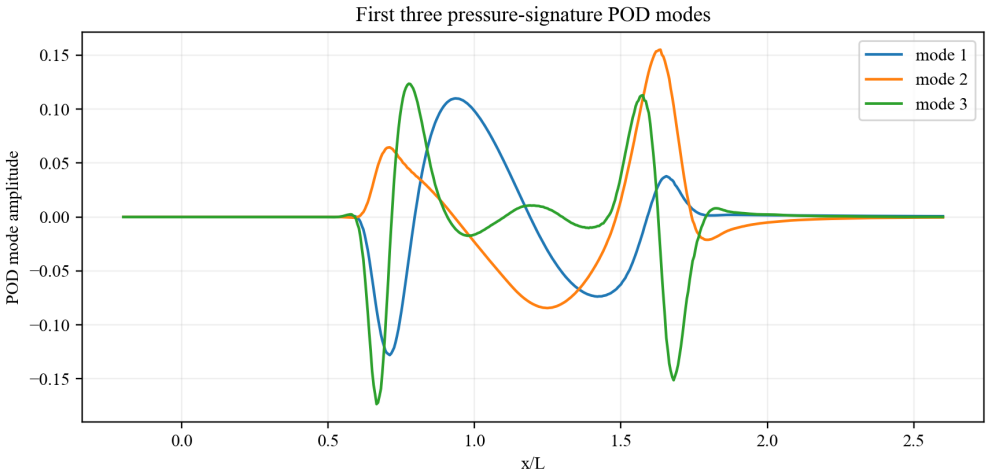


Figure 9: The first three training pressure modes. The modes encode coherent changes in amplitude and wave position across the geometry family.

11 The neural surrogate

The surrogate does not directly predict 561 independent pressure samples. It predicts the compact latent vector plus pressure drag:

$$(a, b) \longrightarrow [z_1, \ldots, z_{12}, \log C_{D,p}]. \tag{22}$$

The released clean-label model uses two tanh hidden layers of 32 neurons each, L-BFGS optimization, $\alpha = 0.01$, maximum 3000 iterations, random seed 16, and a full-SVD PCA/POD basis. The logarithm ensures positive reconstructed drag.

If $\mathbf{z}$ is the predicted modal vector, the pressure signature is reconstructed by the same POD expansion. This separation is useful pedagogically: the neural network learns a low-dimensional geometry-to-coefficient map, while the POD basis stores the spatial structure.

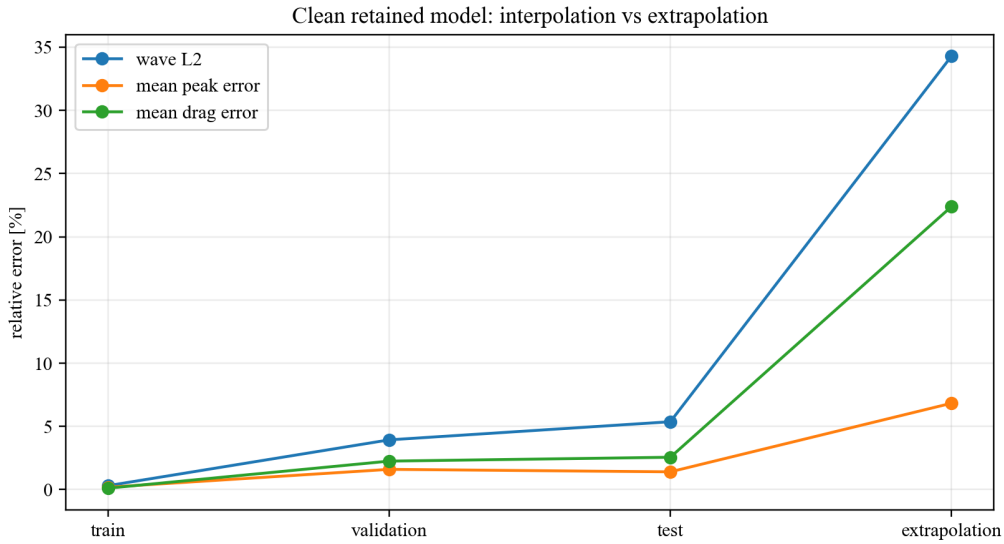


Figure 10: Retained clean-label model errors across the frozen splits. Extrapolation degrades sharply even though interpolation errors remain modest.

The notebook also fits ridge and a fixed-kernel POD–GP on the same 24 standardized training inputs and targets. For the GP,

$$m(X_*) = K(X_*, X)[K(X, X) + 10^{-6} I]^{-1} T, \qquad K(x, x') = \exp[-\|x - x'\|^2/2].$$

The kernel is fixed before validation and test access. Validation selects among the three models; test and extrapolation only assess the selected protocol. GP variance would describe kernel assumptions, not establish CFD uncertainty.

12 Error metrics and what they do not say

Three errors are reported because they answer different questions.

Whole-waveform relative error:

$$\epsilon_{\text{wave}} = \frac{\|\hat{\mathbf{y}} - \mathbf{y}\|_2}{\|\mathbf{y}\|_2}. \tag{23}$$

Mean relative positive-peak error:

$$\epsilon_{\text{peak}} = \left| \frac{\max \hat{C}_p - \max C_p}{\max C_p} \right|. \tag{24}$$

Pressure-drag error:

$$\epsilon_D = \left| \frac{\widehat{C_{D,p}} - C_{D,p}}{C_{D,p}} \right|. \tag{25}$$

A shifted shock can give a modest peak error and a large waveform error. Conversely, aggregate L_2 error can conceal one poor geometry.

For the clean retained model, same-mesh aggregate waveform errors are 0.31% on training, 3.94% on validation, 5.37% on test, and 34.29% on extrapolation. The test worst-case waveform error is 11.78%; extrapolation worst-case error reaches 43.19%. Against the eight finer CFD references, aggregate waveform error is 7.20%, mean peak error 2.24%, mean drag error 2.27%, and worst-case waveform error 15.37%.

Split	Wave L_2	Mean peak	Mean drag	Worst wave
Validation	3.94%	1.60%	2.25%	6.34%
Test	5.37%	1.41%	2.56%	11.78%
Extrapolation	34.29%	6.83%	22.39%	43.19%
Finer CFD test	7.20%	2.24%	2.27%	15.37%

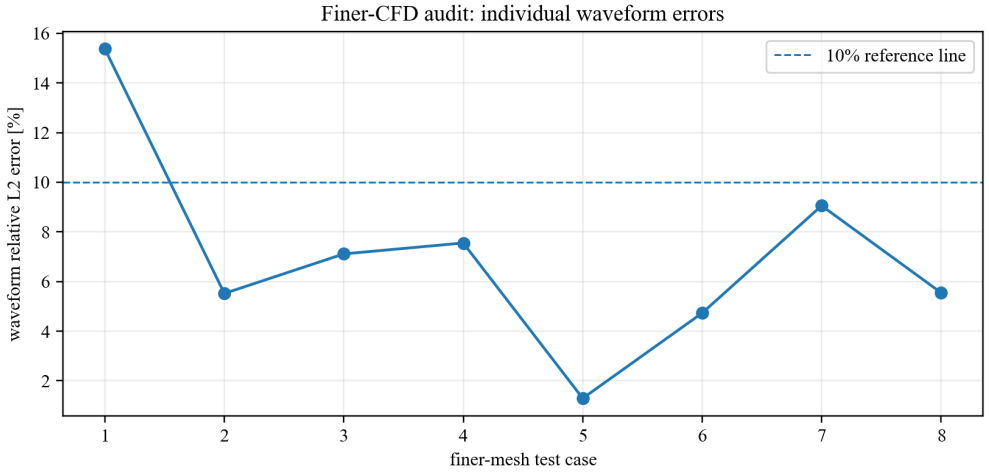


Figure 11: Individual waveform errors against the eight finer CFD references. Aggregate acceptance does not mean every case lies below 10%.

Key idea. The interpolation result is useful within the known two-parameter family. The extrapolation failure is equally important evidence: the model has not learned general supersonic aerodynamics.

13 Constrained shape optimization

The engineering objective is to reduce the maximum positive pressure coefficient on the $r/L = 0.5$ sampling line while limiting pressure drag. The optimization problem is

$$\min_{a,b} \quad J(a,b) = \max_{x} C_p(x;a,b), \tag{26}$$

$$\text{subject to} \quad C_{D,p}(a,b) \leq 1.02\, C_{D,p}^{\text{baseline}}, \tag{27}$$

with volume already enforced by the parameterization.

A derivative-free optimizer is appropriate because the sampled maximum can be nonsmooth and the neural model is cheap to evaluate. But a cheap optimizer may exploit local surrogate errors. Therefore a neural optimum is a *proposal*, not a verified aerodynamic result.

The retained historical candidate has

$$a = 0.3249533, \qquad b = 0.1223779. \tag{28}$$

Its historical surrogate predictions are preserved for provenance, but acceptance is based on fresh SU2 8.0.1 recomputation of the fixed geometry.

Check your understanding. Why is ordinary test-set accuracy not sufficient evidence that an optimizer-selected point is trustworthy?

14 Fresh CFD recomputation of the retained design

The retained baseline and optimized geometries were recomputed at two design mesh levels. At level 1.5,

$$C_{p_{\text{base}}}^{\max} = 0.0333075, \qquad C_{D,p_{\text{base}}} = 0.00135423, \tag{29}$$

$$C_{p_{\text{opt}}}^{\max} = 0.0262504, \qquad C_{D,p_{\text{opt}}} = 0.00130281. \tag{30}$$

This corresponds to a 21.19% peak reduction and a 3.80% drag reduction. At level 2,

$$C_{p_{\text{base}}}^{\max} = 0.0341276, \qquad C_{D,p_{\text{base}}} = 0.00136763, \tag{31}$$

$$C_{p_{\text{opt}}}^{\max} = 0.0270753, \qquad C_{D,p_{\text{opt}}} = 0.00131947, \tag{32}$$

which gives a 20.66% peak reduction and a 3.52% drag reduction.

Observed level-1.5-to-level-2 changes are 2.40% in baseline peak and 0.98% in baseline drag; 3.05% in optimized peak and 1.26% in optimized drag. The optimized waveform L_2 mesh change is 5.39%. The declared peak and drag mesh gates pass; that does not mean the entire waveform is converged to below 5%.

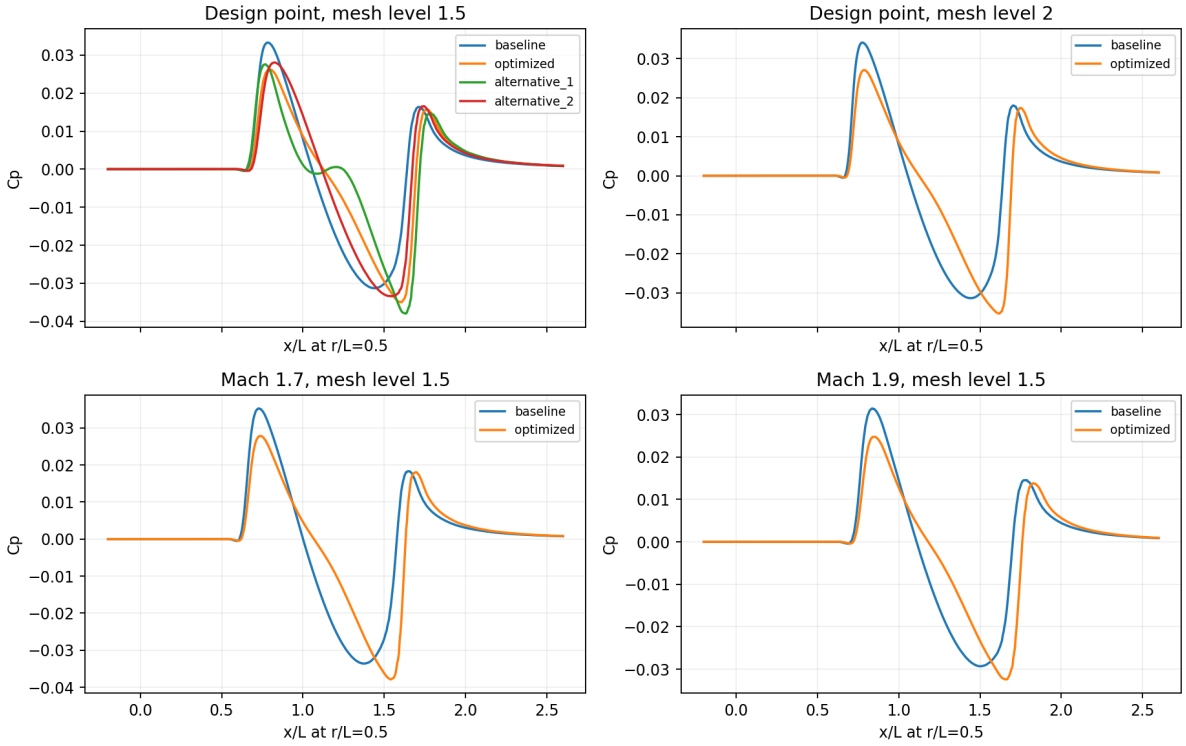


Figure 12: Independent SU2 8.0.1 recomputation of the retained candidate, including design-point mesh comparison and off-design Mach-number checks.

Quantity	Mesh 1.5	Mesh 2
Peak reduction	21.19%	20.66%
Pressure-drag change	-3.80%	-3.52%
Drag-ratio constraint	0.9620	0.9648

15 Off-design robustness at Mach 1.7 and 1.9

The surrogate training campaign fixes Mach number at 1.8. To test the *geometry*, rather than asking the network to extrapolate in Mach number, the baseline and retained candidate were solved directly with CFD at $M = 1.7$ and $M = 1.9$.

At $M = 1.7$, the optimized geometry reduces peak pressure by 21.01% and pressure drag by 3.23%. At $M = 1.9$, the peak reduction is 21.17% and drag reduction is 4.31%. These are direct CFD stress tests of the fixed geometry. They do not establish robustness over a complete flight envelope.

Key idea. A geometry can be tested outside the training operating condition without pretending that the neural network was trained to predict those new conditions.

16 NASA SEEB-ALR: an independent reference, not a successful Week 16 validation

The NASA SEEB-ALR benchmark serves a different purpose from the Taylor–Maccoll cone and from the two-parameter learning body. The public reference includes as-built geometry, wind-tunnel pressure records, coordinate-alignment information, and a NASA-hosted LAVA CFD solution. The comparison condition is $M = 1.6$ at zero incidence.

NASA reports pressure excess normalized by p_∞, while our design-body plots use C_p. Section 3 showed the required conversion. The experimental curves also carry coordinate shifts defined by the original processing macros; those shifts must be preserved rather than tuned to align shocks visually.

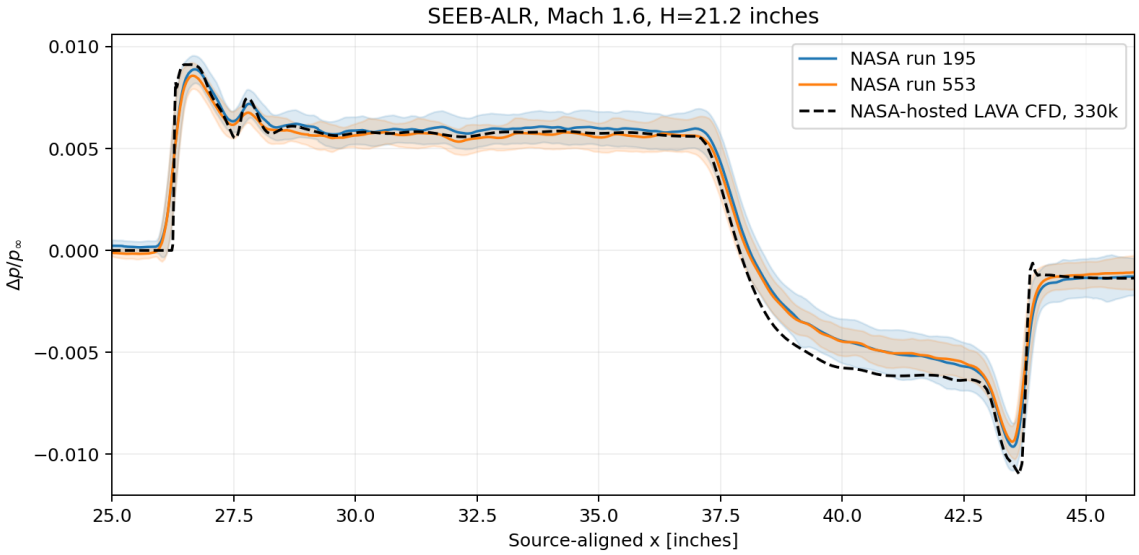


Figure 13: Retained NASA SEEB-ALR reference comparison. The LAVA result is an independent numerical solution, not an exact answer. Experimental agreement and solver-to-solver agreement answer different questions.

17 The failed NASA reproduction is scientifically useful

The attempted SU2 NASA replay did not establish an accepted mesh-refined, physically admissible family. Earlier nose treatment produced unphysical stagnation-region behavior; later recovery runs still failed the declared convergence/physical gates. Those failures are retained rather than hidden.

This is an important lesson for scientific machine learning: a visually plausible pressure trace or a solver exit code is not enough. A reference CFD label must first survive geometry, numerical, and physical checks before it can be used to validate a learned model.

In the release record, NASA is explicitly marked failed/deferred; the accepted Week 16 core rests on the 44-case teaching-body campaign, the Taylor–Maccoll verification, the clean neural model, and the independently recomputed retained design.

Check your understanding. Would agreement between SU2 and LAVA, without comparison to experiment, constitute experimental validation? Why not?

18 What is still required before claiming a quieter ground boom?

A complete low-boom prediction chain would require, at minimum:

1. a realistic aircraft geometry with volume and lift distributions, including wings and propulsion integration;
2. near-field CFD verified for mesh, numerical convergence, and physical consistency;
3. validation against an appropriate measured near-field dataset;
4. transfer of the near-field signature to a validated propagation solver;
5. atmospheric profiles for temperature, humidity, wind, and molecular relaxation;
6. ground-waveform prediction under several operating/atmospheric conditions;
7. calculation of an accepted psychoacoustic metric such as PLdB with its assumptions and uncertainty.

Consequently,

$$\boxed{\text{lower near-field peak} \;\not\Rightarrow\; \text{automatically lower ground PLdB.}} \tag{33}$$

The Week 16 result is intentionally narrower: within the verified axisymmetric Euler teaching problem, the retained design reduces the off-body peak pressure while satisfying the pressure-drag constraint on both design meshes.

19 Reproducibility, provenance, and final takeaways

The clean campaign uses SU2 8.0.1 and contains 44 accepted cases. Model fitting uses only the 24 training geometries, 12 full-SVD POD modes, and a fixed 32–32 tanh MLP. The retained design is not accepted on surrogate prediction alone; it is recomputed with CFD on two meshes and at two nearby Mach numbers. Raw/compact evidence, hashes, and audit reports are retained separately from classroom-friendly arrays.

The scientific workflow to remember is

$$\boxed{\text{physics} \rightarrow \text{verified CFD} \rightarrow \text{POD} \rightarrow \text{surrogate} \rightarrow \text{optimization} \rightarrow \text{fresh CFD verification}}. \tag{34}$$

Five final lessons.

1. A small residual is not the same as an accurate or physically admissible CFD solution.
2. A small average neural error can hide a poor individual waveform.
3. Good interpolation does not imply safe extrapolation.
4. Optimizers exploit model error; recompute the proposed optimum with the governing equations.
5. Near-field pressure shaping and ground sonic-boom loudness are distinct stages of the physics.

Questions for discussion.

1. Why is fixed volume necessary for a fair geometry optimization?
2. What does a POD mode represent physically, and why must it be fitted on the training set only?
3. Why can the peak error be small while waveform L_2 error is large?
4. Which evidence in this module is verification, which is ML validation, and which is experimental reference evidence?
5. What would you add first if the goal changed from “reduce near-field peak” to “minimize ground PLdB”?

References and companion material

1. G. I. Taylor and J. W. Maccoll (1933), "The air pressure on a cone moving at high speeds," *Proceedings of the Royal Society A*, 139, 278–297.
2. Q. Zheng, Y. Liang, Y. Yang, and C. Pan (2026), "Research on low-drag low-boom supersonic transport configuration using an MDO framework and deep learning methods," *Aerospace Science and Technology*, 178, 113218. DOI: 10.1016/j.ast.2026.113218.
3. SU2 governing equations and official solver documentation: https://su2code.github.io/.
4. Gmsh reference manual: https://gmsh.info/doc/texinfo/gmsh.html.
5. NASA/AIAA Sonic Boom Prediction Workshop resources: https://lbpw.larc.nasa.gov/.
6. FlowMLLab Week 16 notebook: notebooks/week16/W16_Supersonic_Shape_Optimization.ipynb.
7. CFD reproduction guide: cases/week16_lowboom/README.md.
8. Numerical evidence and audit reports: results/week16_lowboom/README.md and results/week16_lowboom/reference/.